

Virtualization

Virtualization: An Introduction (T123)

- **Topic:** Virtualization
- **Description:** Exploring the concept of virtualization and its importance in cloud computing.
- **Source:** Based on provided reference materials.

What is Virtualization? (T123)

- Separating hardware from software for improved system efficiency.
- Allows multiple operating systems (guest OS) to run on the same hardware, independent of the host OS.
- Enhances the use of compute engines, networks, and storage.
- Gartner Report (2009): Virtualization was a top strategic technology.

The Virtualization Layer (T123)

- Additional software layer called the **hypervisor** or **Virtual Machine Monitor (VMM)**.
- Manages VMs and allocates resources.
- Responsible for converting real hardware into virtual hardware.
- Enables different OS (e.g., Linux, Windows) to run simultaneously on the same physical machine.

Levels of Virtualization Implementation (T123)

- **Traditional Computer:** Host OS tailored for hardware architecture.
- **After Virtualization:** Guest OS managed by their own applications, running on the same hardware.
- **Key Component:** Virtualization layer (Hypervisor/VMM).

Hypervisor Architecture (T123)

- Supports **hardware-level virtualization** on bare metal.
- Sits directly between physical hardware and the OS.
- Manages CPU, memory, disk, and network interfaces.

Hypervisor and Xen Architecture (T123)

- Hypervisor software sits directly between the physical hardware and its OS.
- This virtualization layer is referred to as either the VMM or the hypervisor.

Hardware Abstraction Level (T123)

- Virtualization performed directly on top of the bare hardware.
- Creates a virtual hardware environment for a VM.
- Manages underlying hardware through virtualization.
- Aims to upgrade hardware utilization rate by multiple users concurrently.
- Example: Xen hypervisor virtualizing x86-based machines.

Operating System Level Virtualization (T123)

- Abstraction layer between traditional OS and user applications.
- Creates isolated containers on a single physical server.
- OS instances utilize the hardware.

Importance in Cloud Computing (T123)

- Enables resource sharing and isolation.
- Supports the creation of virtual clusters dedicated to different tenants.
- Critical for cloud infrastructure and service delivery.

Further Reading (T123)

- Rosenblum, et al. [53,54]
- Smith and Nair [58,59]
- VMware white papers [71,72]
- Xen hypervisor [7,13,42]
- KVM [31]